

Fundamentals of Convexity

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1 Introduction

1.1 Convex Sets

Definition 1. A set $C \subseteq \mathbb{R}^d$ is convex if for all $x, y \in C$ and $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)y \in C.$$

Definition 2. For subset $S \subseteq \mathbb{R}^n$ we can define the span, affine, and convex hull in terms of linear combinations as

$$\begin{aligned} \text{span}(S) &:= \left\{ \sum_i^N a_i x_i \mid x_i \in S \right\}; \\ \text{aff}(S) &:= \left\{ \sum_i^N a_i x_i \mid x_i \in S, \sum_i^N a_i = 1 \right\}; \\ \text{conv}(S) &:= \left\{ \sum_i^N a_i x_i \mid x_i \in S, a_i \geq 0, \sum_i^N a_i = 1 \right\}. \end{aligned}$$

Note $\text{conv}(S) \subseteq \text{aff}(S) \subseteq \text{span}(S)$. Try to visualize these sets for small examples.

Fact 3. The following operations preserve convexity of sets

- Scalar multiplication: $K \rightarrow cK$;
- Addition: $K_1 + K_2$
- Intersection: $\cap_i K_i$
- Affine transform: $K \rightarrow AK + b$

Note that in this last item we do not require A to be invertible. So this shows convexity is preserved by projections (which will be useful later for the relation between sub-level sets and the epigraph).

1.2 Convex Functions

Definition 4. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

We can also define $f : \mathbb{R}^n \rightarrow (\mathbb{R} \cup +\infty)$, in which case we denote the effective domain $\text{dom}(f) := \{x \mid f(x) < \infty\}$ and require the above condition

$$\forall x, y \in \text{dom}(f), \lambda \in [0, 1] : \quad f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

We will use the following extended arithmetic rules for the extended reals: $0 \cdot \infty = 0, \forall s \in \mathbb{R}_{++} : s \cdot \infty = \infty, \forall s \in \mathbb{R} : s + \infty = \infty$.

Exercise 1. Verify that $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all x, y with the extended arithmetic rules given above implies $\text{dom}(f)$ is a convex set.

For convex $K \subseteq \mathbb{R}^n$ we define the indicator function δ_K and the support function h_K as

$$\delta_K(x) := \begin{cases} 0 & x \in K \\ +\infty & \text{otherwise} \end{cases}; \quad h_K(y) := \sup_{x \in K} \langle y, x \rangle.$$

Verify that both of these are convex functions.

Fact 5. The following operations preserve convexity of functions

- Non-negative scalar multiplication: $f \rightarrow cf$;
- Addition: $f_1 + f_2$
- Point-wise supremum: $\sup_i f_i$
- Restriction: $t \rightarrow f((1 - t)x + ty)$ (or more generally a line or subspace).
- Affine transform: $x \rightarrow f(Ax + b)$
- Partial opt: $x \rightarrow \inf_y f(x, y)$

Theorem 6. If f is differentiable, then f is convex iff

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \text{for all } x, y.$$

Proposition 7. For convex f , x is a local minimum iff it is a global minimum.

Proof: Clearly a global minimum is also a local minimum. For the converse, let $\varepsilon > 0$ such that $f(y) \geq f(x)$ for all $\|y - x\|_2 \leq \varepsilon$. Assume for contradiction x is not the global minimum, so $f(x_*) < f(x)$. Then letting $x_\delta := (1 - \delta)x + \delta x_*$, by convexity we have

$$f(x_\delta) \leq (1 - \delta)f(x) + \delta f(x_*) < f(x).$$

But for small enough δ this contradicts local minimality of x . □

Definition 8. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, let the epigraph be

$$\text{epi}(f) := \{(x, t) \mid f(x) \leq t\}$$

i.e. the 'upwards closure' of the graph of f in $\mathbb{R}^{n+1}$.

Theorem 9. $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a closed convex function iff $\text{epi}(f) \subseteq \mathbb{R}^{n+1}$ is a closed convex set.

Proof: First consider f convex, and we want to show $\text{epi}(f)$ is a convex set. Let $(x, t), (y, s) \in \text{epi}(f)$, which by definition means

$$f(x) \leq t, \quad f(y) \leq s.$$

Now consider convex combination $z := (1 - \lambda)x + \lambda y$ for $\lambda \in [0, 1]$. Then by convexity

$$f(z) \leq (1 - \lambda)f(x) + \lambda f(y) \leq (1 - \lambda)t + \lambda s,$$

where the first step was by convexity of f , and in the second step we used that $(x, t), (y, s) \in \text{epi}(f)$. Rewriting this, we have shown

$$(z, (1 - \lambda)t + \lambda s) = (1 - \lambda)(x, t) + \lambda(y, s) \in \text{epi}(f),$$

which shows $\text{epi}(f)$ is a convex set.

For the converse, assume $\text{epi}(f)$ is closed and convex. Then for $(x, f(x)), (y, f(y)) \in \text{epi}(f)$ we have $(z, t) \in \text{epi}(f)$ for $z := (1 - \lambda)x + \lambda y, t := (1 - \lambda)f(x) + \lambda f(y)$, which implies

$$f((1 - \lambda)x + \lambda y) = f(z) \leq t = (1 - \lambda)f(x) + \lambda f(y),$$

verifying convexity of f . □

Lemma 10. For convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and any $t \in \mathbb{R}$, the sub-level set

$$L_t(f) := \{x \in \mathbb{R}^n \mid f(x) \leq t\}$$

is a convex set. Further, if $f(x) = t$ and f is differentiable at x , then the gradient $\nabla f(x)$ gives a supporting hyperplane for the sub-level set L_t at x .

2 Separation Theorems

2.1 Separating Hyperplane Theorem

Theorem 11. For closed convex K and $p \in \mathbb{R}^n$ with $p \notin K$, there is a separating hyperplane $w \in \mathbb{R}^n$ satisfying

$$\max_{x \in K} \langle w, x \rangle < b < \langle w, p \rangle.$$

Proof: The proof plan is simple: let $x_* := \arg \inf_{x \in K} \|x - p\|_2^2$, then we claim the direction $w := x_* - p$ gives a separating hyperplane.

To show this is well-defined, we reduce to the case of compact K . Take any $x_0 \in K$ and consider

$$L := K \cap \{y \in K \mid \|y - p\|_2^2 \leq \|x_0 - p\|_2^2\},$$

which is convex since K is convex and the second set is the sub-level set of convex function $y \rightarrow \|y - p\|_2^2$. Therefore we can assume K is compact and $x_* = \arg \inf_{x \in K} \|x - p\|_2^2$ is attained by the extreme value theorem. We can in fact show that the optimizer x_* is unique, as $\|\cdot\|_2^2$ is strictly convex, but we will not need this for the proof.

Now we claim that $x_* = \arg \min_{x \in K} \|x - p\|_2^2$ iff $\forall x \in K : \langle x - x_*, x_* - p \rangle \geq 0$. Indeed, we can rewrite the function difference

$$\|x - p\|_2^2 - \|x_* - p\|_2^2 = 2\langle x - x_*, x_* - p \rangle + \|x - x_*\|_2^2,$$

as can be verified directly. If the inner product term is ≥ 0 for all x , then since the norm term is always ≥ 0 , we get $\|x - p\|_2^2 \geq \|x_* - p\|_2^2 \forall x \in K$. Conversely, if there exists $x \in K : \langle x - x_*, x_* - p \rangle < 0$, then consider $x_\varepsilon := (1 - \varepsilon)x_* + \varepsilon x$, which is in K by convexity, and note that

$$\|x_\varepsilon - p\|_2^2 - \|x_* - p\|_2^2 = 2\langle x_\varepsilon - x_*, x_* - p \rangle + \|x_\varepsilon - x_*\|_2^2 = \varepsilon\langle x - x_*, x_* - p \rangle + \varepsilon^2\|x - x_*\|_2^2 < 0,$$

for small enough $\varepsilon > 0$, where we used $x_\varepsilon - x_* = \varepsilon(x - x_*)$, and the last step follows as the inner product term is negative and therefore dominates the quadratic term for small enough $\varepsilon > 0$.

Finally, we verify that $w := x_* - p$ gives a separating hyperplane:

$$\langle p - x_*, w \rangle = -\|p - x_*\|_2^2 < 0, \quad \forall x \in K : \langle x - x_*, w \rangle = \langle x - x_*, x_* - p \rangle \geq 0$$

which by rearranging gives

$$\langle p, w \rangle < \langle x_*, w \rangle \leq \inf_{x \in K} \langle x, w \rangle.$$

□

We also get a version for two sets.

Theorem 12. *For closed convex $C, D \subseteq \mathbb{R}^n$, if they are disjoint $C \cap D = \emptyset$ and one of C or D is bounded, then there is a separating hyperplane $(w, b) \in \mathbb{R}^{n+1}$ satisfying*

$$\sup_{x \in C} \langle w, x \rangle < b < \inf_{x \in D} \langle w, x \rangle.$$

Proof: Consider $K = C - D$ and $p = 0$: there is a hyperplane separating C, D iff there is a hyperplane separating 0 from $C - D$. We need the condition that one of C or D is bounded in order for K to be closed. □

Remark 13. *Both the assumptions (closed and bounded) are necessary for strict separation: open convex intervals $(0, 1), (1, 2) \subseteq \mathbb{R}$ are disjoint but cannot be strictly separated; also $\{(x, y) \mid y \leq 0\}$ and the epigraph of the function $1/x$, both in $\mathbb{R}^2$ are disjoint closed convex sets that ‘meet at ∞ ’ and so cannot be strictly separated.*

If we only require separation $\sup_{x \in C} \langle w, x \rangle \leq \inf_{x \in D} \langle w, x \rangle$, then some of these technical conditions can be dropped, see [BV Section 2.5] for further discussion.

Corollary 14. *For closed convex $K \subseteq \mathbb{R}^n$, recall the support function $h_K(w) := \sup_{y \in K} \langle w, y \rangle$ as defined in exercise 1. Show*

$$K = \{x \in \mathbb{R}^n \mid \forall w \in \mathbb{R}^n : \langle w, x \rangle \leq \sup_{y \in K} \langle w, y \rangle = h_K(w)\}.$$

The proof is left as an exercise, but this result is equivalent to the strong duality result proved in theorem 17.

2.2 Supporting Hyperplanes

Definition 15. *A supporting halfspace H of $K \subseteq \mathbb{R}^n$ at $x \in \partial K$ satisfies: (1) $K \subseteq H$; (2) $x \in \partial H$.*

Corollary 16. *For every closed convex K and $x \in \partial K$, there is a supporting hyperplane $H \supseteq K$ such that $x \in \partial H$.*

Proof: See Theorem 3.1.12 in [Nesterov] and page 51 of [Boyd, Vanderberghe]. □

Exercise 2 (Exercise 2.27 in BV). *$K \subseteq \mathbb{R}^n$ closed with non-empty interior, then K is convex iff it has a supporting hyperplane at every point of its boundary.*

3 Strong Duality

Theorem 17. *Closed convex $K \subseteq \mathbb{R}^n$ is the intersection of all containing halfspaces*

$$K \equiv \bigcap_{H \supseteq K} H.$$

Proof: One direction is clear: for $x \in K, H \supseteq K \implies x \in H$, so $K \subseteq \bigcap_{H \supseteq K} H$. For the converse, consider $z \notin K$, then the separating hyperplane theorem 11 gives $w \in \mathbb{R}^n$ such that

$$\langle w, z \rangle > b > \sup_{x \in K} \langle w, x \rangle.$$

So we can consider halfspace $H := \{x \in \mathbb{R}^n \mid \langle w, x \rangle \leq b\}$, which contains K and does not contain z . Therefore $z \notin \bigcap_{H \supseteq K} H$. $\square$

This is actually an equivalent characterization of closed convex sets.

We can also lift this to functions using the epigraph.

Theorem 18. *Closed convex proper $f : \mathbb{R}^n \rightarrow \{\mathbb{R} \cup +\infty\}$ is the supremum of all affine minorants*

$$f(x) = \sup_{f \geq h} h(x).$$

Here proper means f is never $-\infty$ and is not always $+\infty$.

Proof: Recall $\text{epi}(f) \subseteq \mathbb{R}^n$ is a closed convex set by theorem 9, so our plan is to reduce this to strong duality for convex sets shown above. In particular, theorem 17 shows

$$\text{epi}(f) = \bigcap_{H \supseteq \text{epi}(f)} H,$$

where the intersection is over halfspaces containing our convex set $\text{epi}(f)$.

Now note that $f \geq h \iff \text{epi}(f) \subseteq \text{epi}(h)$ as can be verified directly. (Remark: this statement is true for all functions, not just convex functions). Next we can see that $\text{epi}(\sup_i h_i) = \bigcap_i \text{epi}(h_i)$, i.e. the epigraph of the supremum of functions is the intersection of epigraphs. And finally, we can compute the epigraph of affine function $h(x) = \langle w, x \rangle - b$:

$$(x, t) \in \text{epi}(h) \iff \langle w, x \rangle - b \leq t \iff \langle (w, -1), (x, t) \rangle \leq b.$$

Therefore $\text{epi}(h) = H := \{(x, t) \mid \langle (w, -1), (x, t) \rangle \leq b\}$, i.e. epigraphs of affine functions correspond to halfspaces in $\mathbb{R}^{n+1}$ with normal vector of the form $(w, -1)$. Note we could have equivalently defined the halfspace as

$$H = \{(x, t) \mid \langle \lambda(w, -1), (x, t) \rangle \leq \lambda b\}$$

for any $\lambda > 0$, i.e. it is only the sign of the last coordinate that matters.

With these observations we have reduced our proof to the following: theorem 17 gives that $\text{epi}(f)$ is the intersection of a set of containing halfspaces, and we want to show that it is in fact the intersection of just those halfspaces with normal vector (w, z) where $z < 0$. To remove the $z > 0$ halfspaces, we note that $\text{epi}(f)$ is upwards closed, so

$$\sup_{(x, t) \in \text{epi}(f)} \langle (w, z), (x, t) \rangle = \infty,$$

as we can take t arbitrarily large. Therefore all containing halfspaces must have $z \leq 0$.

In order to remove the $z = 0$ halfspaces we will need to use the properness assumption. We claim that there exists some $z < 0$ halfspace; otherwise if all halfspaces have $z = 0$ as the last

coordinate, then either there exists $f(x) = -\infty$, or $f(x) = +\infty$ for all points, contradicting properness. Therefore there exists some containing halfspace $H_0 := \{(x, t) \mid \langle (w_0, -1), (x, t) \rangle \leq b_0\}$ such that $H_0 \supseteq \text{epi}(f)$.

Now we show that for any $(y, s) \notin \text{epi}(f)$ i.e. $s < f(y)$ there exists an affine halfspace with $z < 0$ that separates this point. By theorem 11, there must exist some separating halfspace; if $z < 0$ then we are done; otherwise it is of the form

$$\langle (w, 0), (y, s) \rangle > \sup_{(x, t) \in \text{epi}(f)} \langle (w, 0), (x, t) \rangle.$$

Now consider normal vector $(w_\varepsilon, -\varepsilon) := (1 - \varepsilon)(w, 0) + \varepsilon(w_0, -1)$ and note for small enough ε we still have

$$\langle (w_\varepsilon, -\varepsilon), (y, s) \rangle > \sup_{(x, t) \in \text{epi}(f)} \langle (w_\varepsilon, -\varepsilon), (x, t) \rangle,$$

where we used our containing halfspace $H_0 := \{(x, t) \mid \langle (w_0, -1), (x, t) \rangle \leq b_0\}$ and convexity of the support function $h_{\text{epi}(f)}$ for the right hand side. Therefore this gives a separating hyperplane with $z < 0$, which corresponds to the epigraph of an affine function $f \geq h$ such that $s < h(y)$, which proves the statement. $\square$

The proof above reduces f to its epigraph and then applies strong duality for convex sets. It is valuable to consider what exactly the supporting hyperplanes of $\text{epi}(f)$ correspond to.

4 First Order Definition of Convexity

Recall that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $x \in \text{dom}(f)^\circ$ if the limit

$$\lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t}$$

exists, and further is a linear function of v . Given an inner product, the gradient is defined as

$$\forall v : \quad \langle \nabla f(x), v \rangle = \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t}.$$

Proposition 19. *Let f be differentiable. Then f is convex iff*

$$\forall x, y : \quad f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

We can apply this theory even in the non-differentiable setting.

Definition 20. *For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $x \in \text{dom}(f)$, the set of subgradients $\partial f(x)$ are the elements $g \in \mathbb{R}^n$ satisfying*

$$\forall y \in \mathbb{R}^n : \quad f(y) \geq f(x) + \langle g, y - x \rangle.$$

Note that the set of subgradients is convex: since it is defined by the above inequalities, $g, h \in \partial f(x) \implies (1 - \lambda)g + \lambda h \in \partial f(x)$.

The most important property of convexity, that local minima are global minima, can also be phrased in terms of first-order information:

Theorem 21. *For convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $x = \arg \min_{y \in \mathbb{R}^n} f(y)$ is the global minimum iff $0 \in \partial f(x)$. For differentiable functions, this is equivalent to $\nabla f(x) = 0$.*

Proof: If x is the global minimum, then

$$\forall y \in \mathbb{R}^n : f(y) \geq f(x) = f(x) + \langle 0, y - x \rangle,$$

i.e. $0 \in \partial f(x)$ verifies the definition of subgradient.

Conversely, if $0 \in \partial f(x)$ then we have

$$\forall y \in \mathbb{R}^n : f(y) \geq f(x) + \langle 0, y - x \rangle = f(x),$$

i.e. x is the global minimum. □

Proposition 22. *If f is convex, then for every $x \in \text{dom}(f)^\circ$, there is a subgradient $\partial f(x) \neq \emptyset$.
Conversely, if $\partial f(x) \neq \emptyset$ for every $x \in \text{dom}(f)$, then f is convex.*

Proof: Follows from exercise 2 applied to the epigraph. □

5 Polar Sets and Cones

Definition 23. *For convex $K \subseteq \mathbb{R}^n$, the polar set is*

$$K^\circ := \{y \in \mathbb{R}^n \mid \forall x \in K : \langle y, x \rangle \leq 1\}.$$

Remark 24. *Relate this to the support function as described in exercise 1.*

Note that $0 \in K^\circ$ always; and note that K° is a convex set, even if K is not convex.

Theorem 25. *If $0 \in K$ then $K^{\circ\circ} = K$. More generally,*

$$K^{\circ\circ} = \text{conv}(K \cup \{0\}).$$

Proof: We focus on the case $0 \in \text{int}(K)$ and leave the remainder as an exercise. We first show $K \subseteq K^{\circ\circ}$, so consider $x \in K$; then $w \in K^\circ \implies \langle w, x \rangle \leq 1$ so $\sup_{w \in K^\circ} \langle x, w \rangle \leq 1$, i.e. $x \in (K^\circ)^\circ$.

For the converse, we show the contrapositive. So consider $y \notin K$, then there is a separating hyperplane $\langle w, y \rangle > \sup_{x \in K} \langle w, x \rangle = h_K(w)$. Since $0 \in \text{int}(K)$ we have $h_K(w) > 0$, so $\bar{w} := w/h_K(w) \in K^\circ$. But then this shows $\sup_{v \in K^\circ} \langle v, x \rangle \geq \langle \bar{w}, x \rangle > 1$ i.e. $x \notin (K^\circ)^\circ$. □